PaperForm

Database Architecture Document

Platform: **Supabase (PostgreSQL)**

Schema: **public** | Auth: **auth (Supabase built-in)**

Generated: **September 2026**

Tables: **7** | Functions: **4** | Triggers: **9**

Entity Relationship Diagram

Visual overview of all tables and their relationships within the PaperForm database.

Primary Key Foreign Key Indexed RLS Enabled Trigger

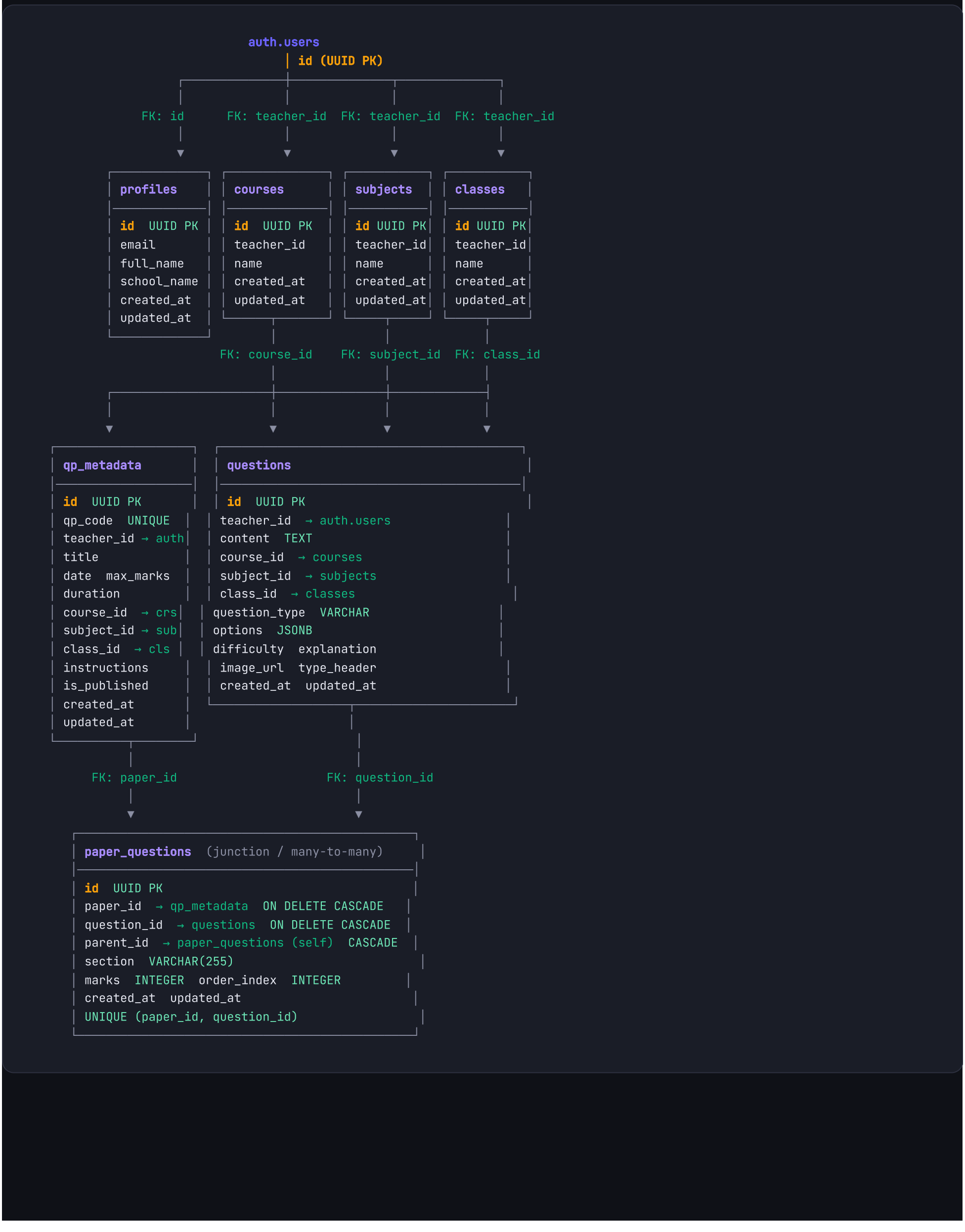


Table Definitions

TABLE 1 **public.profiles**

COLUMN	TYPE	CONSTRAINTS	NOTES
id	UUID	PK, FK → AUTH.USERS	Mirrors auth user
email	VARCHAR(255)	NOT NULL, UNIQUE	
full_name	VARCHAR(255)		
school_name	VARCHAR(255)		
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	Auto-updated via trigger

TABLE 2 **public.courses**

COLUMN	TYPE	CONSTRAINTS	NOTES
id	UUID	PK, DEFAULT GEN_RANDOM_UUID()	
teacher_id	UUID	NOT NULL, FK → AUTH.USERS	CASCADE delete
name	VARCHAR(255)	NOT NULL	UNIQUE per teacher
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

TABLE 3 **public.subjects**

COLUMN	TYPE	CONSTRAINTS	NOTES
id	UUID	PK, DEFAULT GEN_RANDOM_UUID()	
teacher_id	UUID	NOT NULL, FK → AUTH.USERS	CASCADE delete
name	VARCHAR(255)	NOT NULL	UNIQUE per teacher
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

TABLE 4 **public.classes**

COLUMN	TYPE	CONSTRAINTS	NOTES
id	UUID	PK, DEFAULT GEN_RANDOM_UUID()	
teacher_id	UUID	NOT NULL, FK → AUTH.USERS	CASCADE delete
name	VARCHAR(255)	NOT NULL	UNIQUE per teacher
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

TABLE 5 public.qp_metadata

COLUMN	TYPE	CONSTRAINTS	NOTES
id	UUID	PK DEFAULT GEN_RANDOM_UUID()	
qp_code	VARCHAR(10)	NOT NULL UNIQUE	Auto-generated QPXXXX
teacher_id	UUID	NOT NULL FK → AUTH.USERS	CASCADE delete
title	VARCHAR(255)	NOT NULL	
date	DATE		Exam date
max_marks	INTEGER		
duration	INTEGER		In minutes
course_id	UUID	FK → COURSES	SET NULL on delete
subject_id	UUID	FK → SUBJECTS	SET NULL on delete
class_id	UUID	FK → CLASSES	SET NULL on delete
instructions	TEXT		
is_published	BOOLEAN	DEFAULT FALSE	
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

TABLE 6 public.questions

COLUMN	TYPE	CONSTRAINTS	NOTES
id	UUID	PK DEFAULT GEN_RANDOM_UUID()	
teacher_id	UUID	NOT NULL FK → AUTH.USERS	CASCADE delete
content	TEXT	NOT NULL	Question text / HTML
course_id	UUID	FK → COURSES	SET NULL on delete
subject_id	UUID	FK → SUBJECTS	SET NULL on delete
class_id	UUID	FK → CLASSES	SET NULL on delete
question_type	VARCHAR(50)	NOT NULL DEFAULT 'SUBJECTIVE'	mcq / subjective / fillblank / truefalse / match
options	JSONB	DEFAULT '[]'	MCQ choices array
difficulty	VARCHAR(50)	DEFAULT 'MEDIUM'	easy / medium / hard
explanation	TEXT		
image_url	TEXT		
type_header	TEXT		Display grouping header
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

TABLE 7 – JUNCTION

public.paper_questions

COLUMN	TYPE	CONSTRAINTS	NOTES
id	UUID	PK · DEFAULT GEN_RANDOM_UUID()	
paper_id	UUID	NOT NULL · FK → QP_METADATA	CASCADE delete
question_id	UUID	NOT NULL · FK → QUESTIONS	CASCADE delete
parent_id	UUID	FK → PAPER_QUESTIONS (SELF)	Sub-question support; CASCADE delete
section	VARCHAR(255)	NOT NULL · DEFAULT 'A'	Custom section names supported
marks	INTEGER	NOT NULL · DEFAULT 1	
order_index	INTEGER	NOT NULL · DEFAULT 0	Display ordering
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

Indexes

courses

idx_courses_teacher_id (teacher_id)

idx_courses_teacher_name (teacher_id, name) UNIQUE

subjects

idx_subjects_teacher_id (teacher_id)

idx_subjects_teacher_name (teacher_id, name) UNIQUE

classes

idx_classes_teacher_id (teacher_id)

idx_classes_teacher_name (teacher_id, name) UNIQUE

qp_metadata

idx_qp_teacher_id (teacher_id)

idx_qp_code (qp_code)

idx_qp_updated_at (updated_at DESC)

questions

idx_questions_teacher_id (teacher_id)

idx_questions_course_id (course_id)

idx_questions_subject_id (subject_id)

idx_questions_class_id (class_id)

paper_questions

idx_pq_paper_id (paper_id)

idx_pq_question_id (question_id)

idx_pq_section (section)

idx_pq_order (order_index)

UNIQUE (paper_id, question_id)

Row Level Security (RLS)

All tables have RLS enabled. Each table restricts SELECT, INSERT, UPDATE, and DELETE to rows owned by the authenticated user (teacher_id = auth.uid()). The paper_questions table uses a subquery to verify ownership via qp_metadata.

profiles

SELECT → id = auth.uid()

UPDATE → id = auth.uid()

courses / subjects / classes

SELECT → teacher_id = auth.uid()

INSERT → teacher_id = auth.uid()

UPDATE → teacher_id = auth.uid()

DELETE → teacher_id = auth.uid()

qp_metadata / questions

SELECT → teacher_id = auth.uid()

INSERT → teacher_id = auth.uid()

UPDATE → teacher_id = auth.uid()

DELETE → teacher_id = auth.uid()

paper_questions

SELECT → paper_id IN (SELECT id FROM qp_metadata WHERE teacher_id = auth.uid())

INSERT → paper_id IN (...)

UPDATE → paper_id IN (...)

DELETE → paper_id IN (...)

Triggers

Auto-Update Timestamps

BEFORE UPDATE → update_updated_at()

Applied to: profiles, courses, subjects, classes, qp_metadata, questions, paper_questions

Auto-Generate QP Code

BEFORE INSERT ON qp_metadata

→ generate_qp_code()

Format: QPXXXX (4 random digits)

Retries up to 10× on collision

Auto-Create Profile

AFTER INSERT ON auth.users

→ handle_new_user()

SECURITY DEFINER

Reads raw_user_meta_data for name/school

Helper Functions

get_paper_total_marks(p_id UUID)

Returns INTEGER

SUM(marks) from paper_questions

get_paper_questions_count(p_id UUID)

Returns INTEGER

COUNT(*) from paper_questions